LESSON PLAN

FOLG (26)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	05
Total No. of Lesson Plans	9	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Engine Health monitoring by using Debris analysis
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Engine Health monitoring by using Debris analysis.
5. Specific Objectives	26.1.1.4 Trainee will be able to explain the Debris Analysis EHM Technique

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of Debris Analysis Case Studies; Safety: Handle simulated debris samples with care; Instructor supervision required during sample identification.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary purpose of Engine Health Monitoring?
(b) Name two common methods used for EHM.
(c) Define the term 'Debris Analysis' in the context of EHM.

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Introduction to Debris Analysis EHM • Sources of Debris in Engine Systems • Types of Debris Particles • Collection Methods for Debris Samples • Initial Screening Procedures	L & Ds	PPT, IB	15 min
(b)	• Laboratory Analysis Techniques • Microscopic Examination Methods • Elemental Composition Analysis • Interpretation of Debris Patterns • Correlation with Engine Performance	L & Ds	PPT, PDF	20 min
(c)	• Reporting Debris Analysis Results • Maintenance Actions Based on Findings • Limitations of Debris Analysis • Integration with Other EHM Techniques • Case Study Review	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the step-by-step process of Debris Analysis.
(b) How does debris analysis help in predicting engine failures?
(c) What are the limitations of relying solely on debris analysis?

12. Home Assignment and Follow Up	Prepare a brief report on how Debris Analysis complements Gas Path Analysis for comprehensive Engine Health Monitoring (Bridge to LP 06).

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Engine Health monitoring by using Gas path analysis	Refer to the master syllabus reference list	LP 06

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min